Allison Newman

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Education

Cornell University, Bachelor of Science, Mechanical Engineering GPA: 3.93 | Aug. 2024 – Exp. Dec. 2027
Cornell University, Planned Master of Engineering, Aerospace Engineering Jan. 2028 – May 2028
Relevant courses: Statics, Mechanics of Materials, Dynamics, Fluid Mechanics (TA), Heat Transfer, Thermodynamics, Intro. to Aeronautics, Spaceflight Mechanics

Experience

Lockheed Martin RMS–Sikorsky, Assembly & Flight Operations, Mechanical Eng. Intern May – Jul. 2026
- Designed structural fixtures in CATIA for AFSD substrate marking, LPBF part removal, and welding-stock deposition, preventing part deformation and feed misalignment that could compromise 72-hour builds.
- Modeled inert-gas purging for AFSD using STAR-CCM+ and Ansys, implementing steady-state, transient, turbulent-flow, and passive-scalar models to identify process inefficiencies and recommend system improvements.
- Used GrabCAD Print and Insight to slice, print, and package 26 ASA parts of the modernized Black Hawk nose for a trade-show in under 2 weeks utilizing 3 F900 printers.
- Developed an 18 hp hybrid VTOL propulsion concept in Creo targeting 20-minute hover endurance; conducted engine–motor trade studies, selected rotors and ESCs, and developed ground/flight test procedures, earning 2nd place in a Sikorsky engineering review.

Cornell University Unmanned Air Systems (CUAir), Structures and Payloads Engineer Nov. 2024 – Present
- Designing, manufacturing, and testing a carbon-fiber composite autonomous aircraft for the CUASC competition; performing fuselage and bulkhead layups and contributing to a 36% aircraft weight reduction to meet the revised 35 lb limit (previously 55 lb).
- Placed 2nd in flight and design at CUASC UAV 2025 international competition.
- Competed at SUAS 2026, providing quick & scrappy weight savings ideas to pass safety test. Reassembled vertical propulsion system in under 2 hours after a crash, enabling team to pass second safety test to become flight viable again.
- Prototyping a novel plane-recovery system to prevent timeline setbacks of crashes. Weighing ballistic launch vs. pneumatic systems.

Cornell University Intro Fluid Mechanics, Undergraduate Teaching Assistant Aug. 2026 – Present
- Demonstrating fluid concepts using Schlieren imaging, forklift hydraulics, and engine dissections to 60 3rd-year students weekly.

Projects

Tilt Rotor Mechanism for 35 lb eVTOL Aircraft, CUAir Project
- Designed and sized a tilt-rotor mechanism delivering 45 lb of horizontal thrust at 25 m/s, enabling flight-mode transitions at 180°/s with 1.5× the required flight thrust and a 1.4 factor of safety against bracket yielding due to coupled moment and distributed load.
- Designed 5052 aluminum rotating motor brackets and servo mounts, PETG boom adapter block, and axle–collar–servo shaft-extension hardware stackup in SolidWorks, integrating actuation, structural support, and rotational interfaces.
- Calculated load paths, bending, bearing, buckling, and yield margins using first-principles statics and dynamics; evaluated structural stiffness and deflection in Ansys Static Structural FEA.
- Analyzed crash loads using custom MATLAB rigid-body dynamics calculations and Ansys FEA, identifying a 50 N · m bracket-yield threshold and gear-disengagement risk; designed a steel retaining plate to increase predicted rotational stiffness by 8×.
- Developed and executed coupled thrust-and-rotation tests across 0–180° rotation and 0–100% thrust, verifying boom deflection remained within structural limits and inspecting gear teeth for chipping.
- Monitored 5052 aluminum bracket temperatures during ground testing; integrated Garolite thermal insulation to maintain temperatures below a 100°C design limit, mitigating a predicted 6% reduction in elastic stiffness.
- Collaborated with electrical subteam to resolve a 35% RPM measurement discrepancy, identifying incorrect manufacturer-specified motor pole-pair count (19 vs. 14) and validating motor performance against specifications.

Object Collecting Robot, Mechatronics Project
- Ranked 3rd of 60 teams in autonomous block-collection competition, achieving 15 blocks/min using a custom-designed collection mechanism.
- Designed in SolidWorks using COTS + 9 bespoke parts fabricated from acrylic/ PLA; used laser cutting and rotary tools.
- Programmed Arduino in C++ and registers. Utilized 3 servos and 2 QTIs + color sensor for board localization/border detection.

Affiliations

Society of Women Engineers, Communications Codirector Sep. 2024 – Present
- Contacted 20+ professors at Cornell to secure a diverse set of mentors for the annual student-faculty dinner with 30+ attendees.

Academic Excellence Workshop, Facilitator for MATH2940 – Linear Algebra Aug. 2025 – May 2026
- Developed and facilitated 2-hour weekly workshops with co-facilitator to reinforce key Linear Algebra principles for undergraduate students; drafted, proofread, and tested worksheets and games to engage students while highlighting key concepts, totaling 4+ hours of prep weekly.

Skills and Certifications

Technical: Rapid Prototyping & End-to-end Project Ownership, SolidWorks (Sheet Metal), CATIA, 3DX, Ansys FEA (Static Structural), STAR-CCM+ CFD, GD&T, DFMA, Hand Calculations (Statics/Dynamics), Carbon Fiber Composite Layups & Testing, FDM 3D Printing (Insight, GrabCAD), Machining (Mill, Lathe, AFSD Welding), MATLAB, Arduino (C++), Java
Operations & Safety: Solumina, Confluence, Project Management, OSHA Safety, ESH, FOD Awareness